AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
MECHATRONICS ENGINEERING DEPARTMENT

Spring, 2026

MCT344-342: Industrial Robotics

Sheet 03: Forward Kinematics for Robotics

1) Consider the three-link planar manipulator shown in **Figure 1**. Derive the forward kinematic equations using the DH convention.

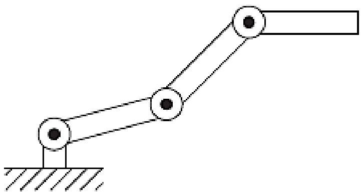

Figure 1 Three-link planar arm

2) Consider the two-link Cartesian manipulator shown in **Figure 2**. Derive the forward kinematic equations using the DH convention.

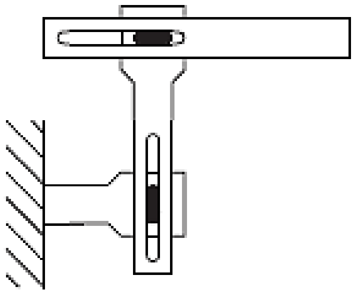

Figure 2 Two-link Cartesian robot

3) Consider the two-link manipulator of **Figure 3**, which has joint 1 revolute and joint 2 prismatic. Derive the forward kinematic equations using the DH convention.

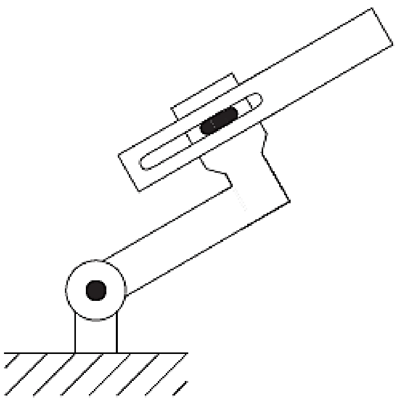

Figure 3 Two-link planar arm

4) Consider the three-link planar manipulator of **Figure 4**. Derive the forward kinematic equations using the DH convention.

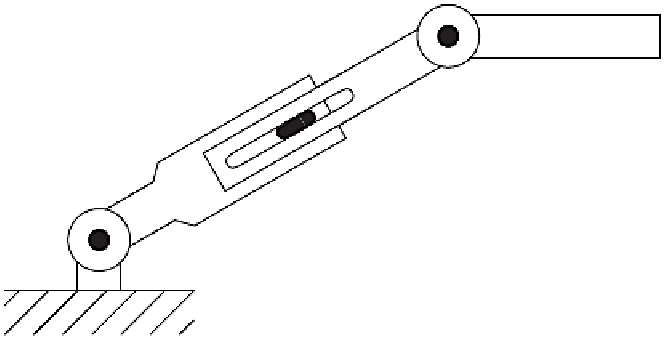

Figure 4 Three-link planar arm with prismatic joint

5) Consider the three-link articulated robot of **Figure 5**. Derive the forward kinematic equations using the DH convention.

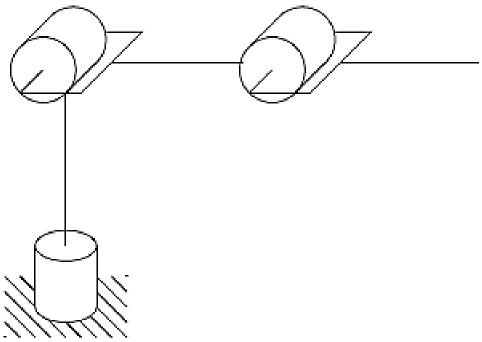

Figure 5 Three-link articulated robot

6) Consider the three-link Cartesian manipulator of **Figure 6**. Derive the forward kinematic equations using the DH convention.

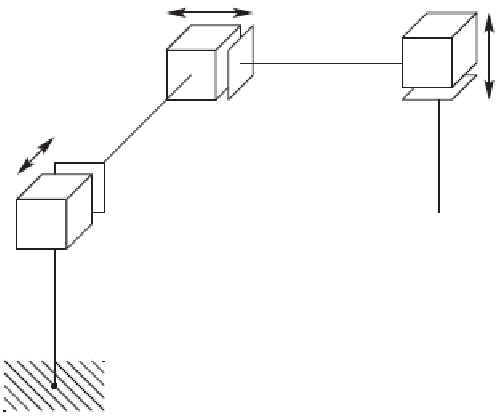

Figure 6 Three-link Cartesian robot.

7) Attach a spherical wrist to the three-link articulated manipulator of Problem 5 as shown in **Figure 7.** Derive the forward kinematic equations for this manipulator.

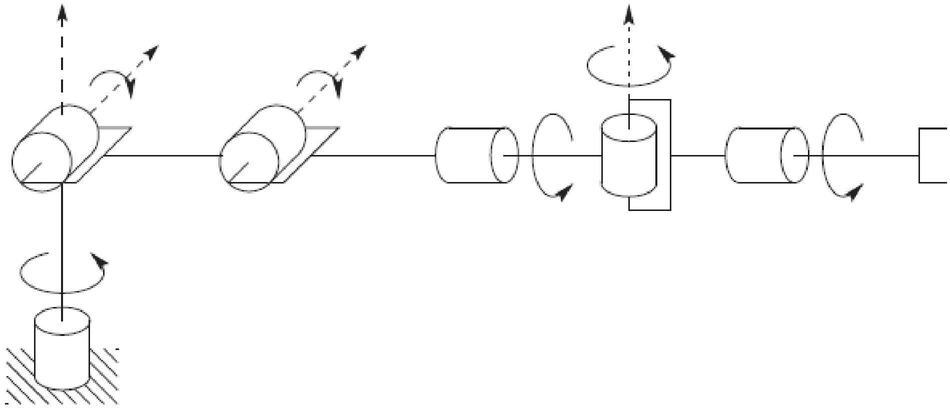

Figure 7 Elbow manipulator with spherical wrist.

8) Attach a spherical wrist to the three-link Cartesian manipulator of Problem 6 as shown in **Figure 8**. Derive the forward kinematic equations for this manipulator.

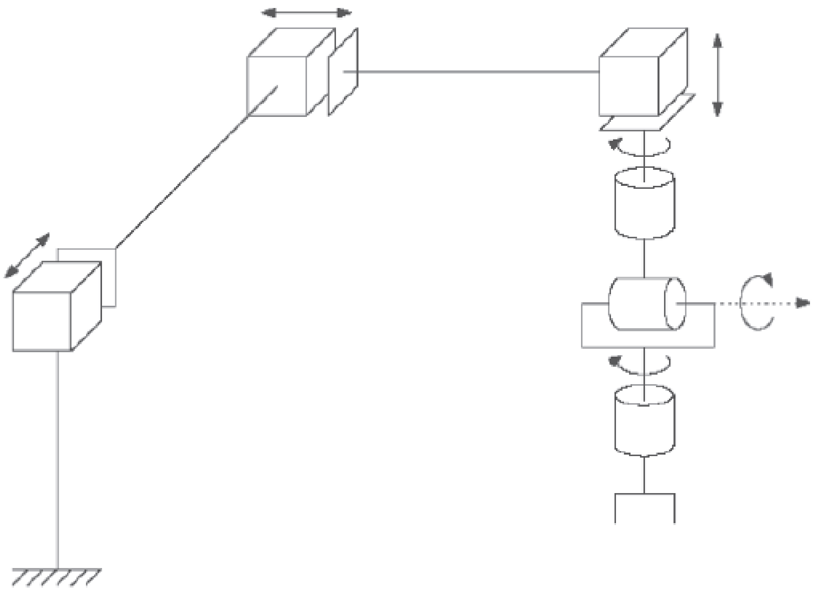

Figure 8 Cartesian manipulator with spherical wrist.

9) Consider the ***PUMA 260*** manipulator shown in **Figure 9**. Derive the complete set of forward kinematic equations by establishing appropriate DH coordinate frames, constructing a table of DH parameters, forming the A matrices, etc.

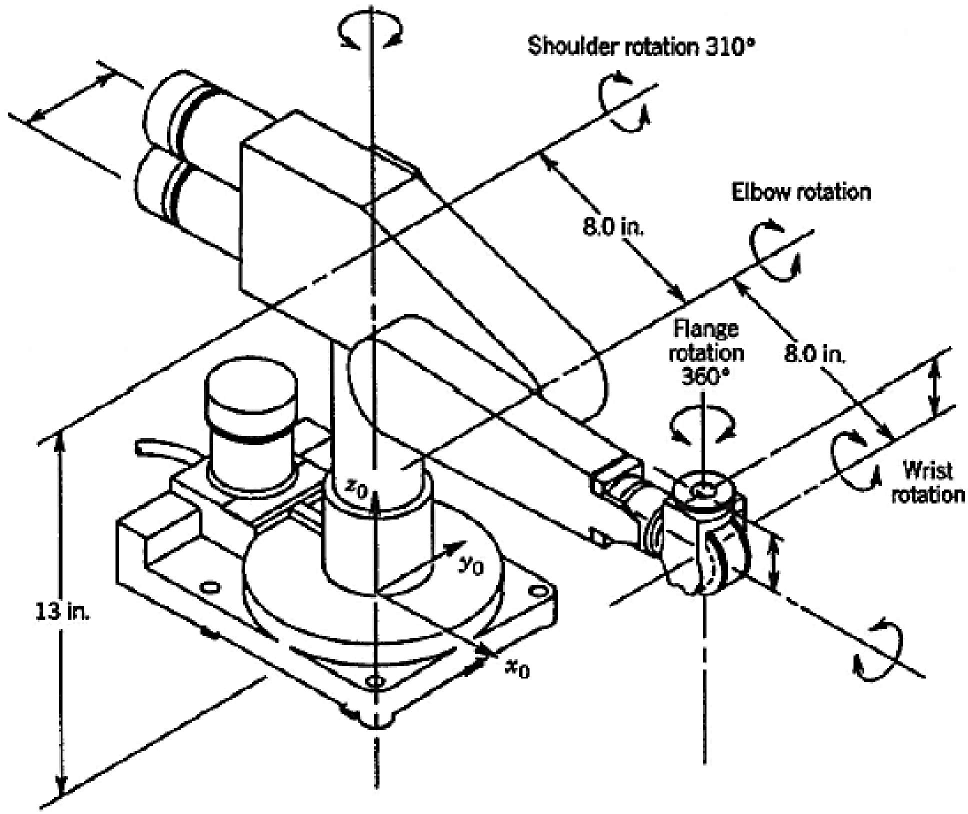

Figure 9 PUMA 260 manipulator.